



Governing digital innovation ecosystems under risk and uncertainty: IP circulation, platform dependency, and policy orchestration in Korean content industries

Taejun (David) Lee

KDI School of Public Policy and Management, 263 Namsejong-ro, Sejong-si, 30149, Republic of Korea

ARTICLE INFO

Keywords:

Digital innovation
Risk and uncertainty
Innovation ecosystems
Intellectual property (IP) circulation
Platform dependency
Policy orchestration
Korean content industry

ABSTRACT

Digital innovation in cultural and creative industries unfolds under compounded risk and uncertainty. Drawing on Knight's (1921) distinction, ecosystem actors face both calculable "known unknowns," such as production-cost volatility, project financing risk, and platform-competition dynamics that can in principle be pooled or diversified, and irreducible "unknown unknowns," such as unpredictable audience reception, generative artificial intelligence (AI) disruption, platform restructuring, and contested intellectual property (IP) appropriation, which cannot be reduced to probabilities. Existing innovation ecosystem theories, grounded largely in manufacturing and technology-driven contexts, struggle to explain how value creation, coordination, and governance operate in industries centered on intangible and symbolic assets under platform dependency. Addressing this gap, this study develops a conceptual framework for digital content innovation ecosystems under conditions of high risk and high uncertainty. Methodologically, it is a conceptual, theory-synthesizing contribution: the framework is developed through integrative synthesis of innovation ecosystem theory, platform studies, and cultural economics, and is then illustrated through an interpretive reading of the Korean content industry between 2023 and 2026, where IP, digital platforms, and policy institutions interact under intense platform dependency and AI-driven transformation. The framework specifies three generative mechanisms, namely IP circulation, platform coordination, and network integration, and positions policy orchestration as a higher-order governance mechanism that operates across them by pooling calculable risk through policy finance and by absorbing irreducible uncertainty through adaptive, iterative coordination. The analysis further theorizes a central contingency: whether IP circulation produces cumulative ecosystem value or reproduces dependency hinges on the allocation of IP ownership. Rather than demonstrating stabilized outcomes, the study characterizes the instruments and attempts through which ecosystem-level orchestration seeks to manage unknowns, thereby extending innovation ecosystem theory to platform-mediated cultural industries.

1. Introduction

The rapid diffusion of digital technologies has fundamentally transformed the structure and dynamics of cultural and creative industries, intensifying uncertainty, volatility, and platform dependency within contemporary innovation ecosystems. In particular, the emergence of digital platforms, over-the-top (OTT) services, algorithmic recommendation systems, and data-driven distribution infrastructures has reconfigured how content is produced, circulated, and consumed across global markets. Unlike traditional industrial systems characterized by vertically integrated firms and relatively stable value chains, contemporary content industries are increasingly organized around platform-

mediated ecosystems in which value creation depends on distributed coordination among creators, intermediaries, digital infrastructures, and global audiences (Nieborg & Poell, 2018; Parker et al., 2016; Srnicek, 2017; Van Dijck et al., 2018). While these transformations have enabled scalability, rapid internationalization, and new forms of collaborative innovation, they have also generated increasing asymmetries in visibility, monetization, and market access under conditions of intensified uncertainty and ecosystem fragmentation (Gillespie, 2018; Nambisan et al., 2019; Zuboff, 2019).

Empirically, the study draws on the Korean content industry as an illustrative ecosystem in which the interaction among IP, platforms, policy institutions, and transnational market expansion has become

E-mail address: davidtjlee@gmail.com.

<https://doi.org/10.1016/j.techsoc.2026.103495>

Received 13 May 2026; Received in revised form 10 July 2026; Accepted 2 August 2026

Available online 3 August 2026

0160-791X/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

particularly visible under rapid digital transformation. Rather than treating Korea as an exceptional case, the analysis employs analytical generalization to clarify how the proposed mechanisms operate within a real-world ecosystem characterized by platform dependency, strong global integration, and active policy intervention (Eisenhardt & Graebner, 2007; Yin, 2018).

Overall, this study makes four interrelated contributions. First, it extends innovation ecosystem theory beyond its technological and manufacturing foundations by incorporating the distinctive characteristics of content industries, particularly the centrality of intangible assets, symbolic value, and platform-mediated coordination. Second, it introduces IP circulation as a recursive generative mechanism of value creation under uncertainty-intensive digital innovation, while theorizing its central contingency: the same circulation dynamics generate cumulative value when originators retain IP ownership but reproduce dependency when ownership is transferred to dominant platforms. Third, it reconceptualizes platforms not as neutral intermediaries but as infrastructural governors whose coordination is analytically distinct from the horizontal, peer-based coordination captured by network integration. Fourth, it advances policy orchestration as a higher-order governance mechanism, extending the orchestration construct from innovation-network and governance research (Abbott & Snidal, 2010; Dhanaraj & Parkhe, 2006) to platform-mediated cultural industries operating under compound risk and uncertainty. Together, these contributions provide a process-oriented, system-level framework that specifies three generative mechanisms, IP circulation, platform coordination, and network integration, coordinated by a higher-order layer of policy orchestration. In doing so, the study engages emerging debates on how ecosystems can manage digital innovation when neither calculable risks (the known unknowns of production-cost volatility, project financing, and platform competition) nor deeper uncertainties (the unknown unknowns of unpredictable audience demand, generative AI disruption, platform restructuring, and contested IP appropriation) can be eliminated through firm-level strategy alone.

2. Literature review

2.1. Innovation ecosystems

The concept of innovation ecosystems has emerged as a central framework for understanding how innovation is generated through interdependent networks of actors rather than within isolated firms. Early innovation theories primarily focused on firm-level capabilities, emphasizing internal resources, technological competencies, and competitive advantage as key drivers of innovation (Dosi, 1982; Nelson & Winter, 1982; Teece, 1986). However, as technological complexity increased and innovation processes became more distributed, scholars increasingly recognized that value creation depends on the coordinated activities of multiple actors operating across organizational and institutional boundaries (Kogut, 2000; Powell et al., 1996). More recently, digital transformation has further intensified interdependence, uncertainty, and coordination complexity within innovation systems, making ecosystem-based perspectives increasingly important for understanding contemporary innovation dynamics (Nambisan et al., 2017; Nambisan et al., 2019; Teece et al., 2016).

A key implication of this perspective is the shift from a firm-centric to a network-centric understanding of innovation. Whereas traditional models assume that firms are the primary loci of innovation, ecosystem approaches highlight the importance of networks, platforms, and institutional arrangements in shaping innovation outcomes (Adner & Kapoor, 2010; Autio & Thomas, 2014; Granovetter, 1985; Powell et al., 1996). Networks facilitate knowledge exchange, resource sharing, and collaborative problem-solving, thereby enabling innovation processes that exceed the capabilities of individual firms. In particular, digital platforms have reinforced the networked nature of innovation by lowering coordination costs and enabling large-scale interactions among

heterogeneous actors across geographically dispersed markets (Gawer & Cusumano, 2014; Kenney & Zysman, 2016; Parker et al., 2016).

In response, this study adopts an ecosystem perspective that foregrounds network-based coordination while recognizing the evolving roles of firms, platforms, and institutions in shaping adaptive ecosystem dynamics. Recent systematic reviews have similarly emphasized that digital technologies are reshaping the structure, function, and evolution of innovation ecosystems and that ecosystem-level analyses must increasingly account for risk and crisis-driven reconfiguration (Brem et al., 2024; Khizar et al., 2025). Building on these developments, the present study seeks to extend innovation ecosystem theory to better account for the distinctive characteristics of digital content industries, where the boundaries between production, distribution, and consumption are increasingly blurred, and where innovation emerges through the circulation of ideas, assets, and interactions across interconnected digital networks. In particular, this approach conceptualizes innovation not as a firm-bound activity, but as a system-level process driven by interdependent mechanisms of coordination, circulation, orchestration, and adaptive governance under conditions of platform dependency and uncertainty.

2.2. Platform economy and digital transformation

The rise of digital platforms has fundamentally reshaped the organization of industries and the processes through which innovation is generated, coordinated, and distributed. Platforms are not merely technological infrastructures, but organizational and institutional forms that facilitate interactions among multiple groups of users, enabling value creation through network effects, algorithmic coordination, and data-driven governance (Evans & Schmalensee, 2016; Parker et al., 2016; Rochet & Tirole, 2003). As such, platforms have emerged as central actors within contemporary innovation ecosystems, redefining the boundaries of firms, markets, and industries (Gawer, 2021; Kenney & Zysman, 2016). At the same time, however, the growing dependence on digital infrastructures has intensified uncertainty, asymmetry, and governance complexity within ecosystem environments, making platform-mediated coordination both a source of innovation and a source of systemic vulnerability (Nambisan et al., 2019; Srnicek, 2017).

Within cultural and creative industries, these transformations are particularly pronounced. Digital platforms such as streaming services, social media platforms, and content-sharing infrastructures have become dominant channels through which cultural products are produced, circulated, and monetized (Cunningham & Craig, 2019; Nieborg & Poell, 2018). These platforms do not merely mediate access to content; they actively shape the conditions of cultural production by determining visibility, discoverability, monetization models, and audience engagement through algorithmic governance and data analytics (Gillespie, 2018; Poell et al., 2022). Consequently, the boundaries between production, distribution, and consumption have become increasingly blurred, giving rise to participatory, co-creative, and data-driven forms of cultural production and circulation (Jenkins, 2006).

At the same time, platformization generates new forms of dependency and asymmetry within innovation ecosystems. While platforms enable global reach, scalability, and low-cost participation, they also concentrate infrastructural power in the hands of a small number of dominant actors that control data, algorithms, standards, and user access (Srnicek, 2017; Van Dijck et al., 2018; Zuboff, 2019). This creates persistent tensions between openness and control, participation and dependency, and innovation and vulnerability. Ecosystem participants must therefore navigate highly uneven governance environments in which platform owners exercise substantial influence over visibility, revenue distribution, and market participation through opaque and constantly evolving governance mechanisms (Gawer & Cusumano, 2014; Tiwana, 2014). Under these conditions, uncertainty becomes an intrinsic feature of digital innovation ecosystems rather than an external disruption.

2.3. Cultural and creative industries

Cultural and creative industries (CCIs), including film, television, music, games, and digital content, exhibit distinctive characteristics that challenge conventional theories of innovation and industrial organization. Unlike traditional manufacturing sectors, where value is primarily derived from physical production, technological efficiency, and standardized outputs, CCIs are fundamentally organized around the creation, circulation, and monetization of intangible, symbolic, and experiential goods (Caves, 2000; Hesmondhalgh, 2019; Throsby, 2001). These industries are characterized by high levels of uncertainty, demand unpredictability, and strong dependence on creative labor and audience engagement, making innovation processes highly non-linear, recursive, and difficult to standardize (DeFillippi et al., 2007; Lampel et al., 2000). As digital transformation accelerates the speed of cultural circulation and intensifies platform dependency, uncertainty has become an increasingly constitutive condition of innovation within digital content ecosystems.

One of the defining features of CCIs is the pervasive uncertainty associated with both production and consumption. As Caves (2000) famously argued, creative products operate according to a “nobody knows” principle, whereby the commercial and cultural success of content cannot be reliably predicted *ex ante*. In Knight's (1921) terms, audience demand in cultural markets is the paradigmatic case of irreducible uncertainty rather than calculable risk, because it cannot be assigned stable probabilities from prior experience. This uncertainty shapes investment decisions, production strategies, and ecosystem structures, often resulting in project-based organizational forms and temporary, flexible collaborations among heterogeneous actors (DeFillippi & Arthur, 1998; Grabher, 2002). Consequently, innovation in CCIs does not follow stable trajectories of technological development or linear research and development processes, but instead emerges through iterative experimentation, recombination of ideas, audience feedback, and adaptive interaction across interconnected networks (Hirsch, 1972; Lampel et al., 2000). Under such conditions, ecosystem actors must continuously coordinate and adapt to volatile market environments, rapidly changing consumer preferences, and shifting platform governance structures.

Closely related to this uncertainty is the immaterial and symbolic nature of cultural goods. Content products are generally non-rival, highly reproducible, and scalable across multiple channels and markets at relatively low marginal cost (Shapiro & Varian, 1999; Towse, 2010). However, their value is not intrinsic or functionally determined, but socially constructed through meaning, interpretation, visibility, and audience reception (Hesmondhalgh, 2019; Throsby, 2001). This combination of scalability and symbolic value creates distinctive dynamics of value creation and capture in which economic outcomes depend heavily on reputation, cultural legitimacy, algorithmic visibility, and network effects rather than purely technological performance (Cunningham & Potts, 2015; Potts et al., 2008). As a result, conventional innovation models centered on efficiency, pricing, and technological complementarities remain insufficient for explaining ecosystem dynamics in digital cultural industries.

The increasing platformization of CCIs further intensifies these dynamics. Digital platforms such as streaming services, social media infrastructures, and content-sharing ecosystems now play a central role in shaping how cultural products are circulated, discovered, monetized, and consumed (Cunningham & Craig, 2019; Nieborg & Poell, 2018). Platforms do not merely distribute content; they actively govern ecosystem interactions through algorithms, data analytics, recommendation systems, and monetization architectures that influence visibility and audience attention (Gillespie, 2018; Poell et al., 2022). Consequently, innovation in CCIs becomes increasingly dependent on infrastructural systems controlled by dominant platform actors, creating new forms of asymmetry, dependency, and governance complexity within digital ecosystems (Srnicke, 2017; Zuboff, 2019). Ecosystem

participants must therefore navigate conditions of heightened uncertainty in which platform rules, algorithmic visibility, and audience attention are continuously evolving and only partially transparent.

2.4. Intellectual property and value creation

Intellectual property (IP) has become a central mechanism of value creation in cultural and creative industries, particularly under conditions of digital transformation, platformization, and uncertainty-intensive innovation. Unlike traditional industrial sectors, where value is primarily embedded in physical products or production processes, digital content industries derive value from intangible assets that can be reproduced, recombined, adapted, and circulated across multiple formats, platforms, and markets (Shapiro & Varian, 1999; Towse, 2010). Under increasingly volatile and platform-mediated market environments, IP functions not merely as a legal construct for protecting ownership, but as a strategic and adaptive resource that enables the generation, coordination, and stabilization of value within innovation ecosystems (Gans et al., 2002; Pisano, 2006; Teece, 1986). This perspective shifts the analytical focus from IP as a mechanism of appropriation toward IP as a dynamic driver of ecosystem coordination, resilience, and recursive value creation.

Several adjacent concepts describe the movement of cultural content across formats and markets, yet each leaves a distinct analytical gap. Transmedia storytelling foregrounds the narrative and audience-experience logic of dispersing a story across media (Jenkins, 2006); the one-source multi-use (OSMU) concept, long established in Korean policy and industry discourse, foregrounds the commercial reuse of a single source across product lines; and the notion of the contingent cultural commodity captures how platformization renders cultural goods continuously reformatted and market-contingent (Nieborg & Poell, 2018). These concepts, however, remain oriented toward the content artifact or the individual firm rather than the ecosystem, and they treat circulation largely as an outcome of authorial or commercial strategy. What remains underdeveloped is a mechanism-level account of how the recursive movement of IP structures value creation, coordination, and adaptation at the level of the ecosystem itself, and of the conditions under which such circulation accumulates value rather than reproducing dependency. Building on this observation, Section 3 develops the concept of IP circulation, which differs from transmedia storytelling and OSMU in definition (an ecosystem-level mechanism rather than a narrative or commercial practice), in unit of analysis (the ecosystem rather than the text or the firm), and in mechanism (the conditions under which recursive circulation accumulates value versus reproduces dependency).

2.5. Research gap

Three interrelated gaps emerge from the preceding review. First, although innovation ecosystem research has successfully shifted attention from firm-centric to network-based explanations of value creation, the literature remains predominantly grounded in technological and manufacturing contexts (Adner, 2017; Jacobides et al., 2018). Existing frameworks tend to emphasize technological complementarities, modular architectures, and supply-side coordination, while paying comparatively limited attention to industries where value creation depends primarily on intangible assets, symbolic meanings, audience engagement, and recursive cultural circulation. While cultural and creative industries research has emphasized uncertainty, symbolic production, and project-based collaboration (Caves, 2000; Lampel et al., 2000; Potts et al., 2008), these insights have largely developed separately from mainstream ecosystem theory. As a result, current ecosystem frameworks remain insufficient for explaining how innovation is coordinated and sustained under conditions characterized by platform dependency, audience volatility, algorithmic governance, and uncertainty-intensive value creation. This reveals a broader limitation

in existing innovation theory, which continues to rely heavily on assumptions of relatively stable architectures and predictable coordination environments.

Second, despite the growing importance of intellectual property in digital cultural production, the mechanisms through which IP drives innovation at the ecosystem level remain conceptually underdeveloped. Existing research has largely treated IP as a legal or economic asset associated with ownership protection and value appropriation (Teece, 1986; Towse, 2010), rather than as a dynamic and adaptive mechanism that structures interactions, circulation, and recombination across interconnected ecosystems. As noted above, practices such as OSMU and transmedia storytelling demonstrate that innovation in digital content industries increasingly emerges through recursive and cumulative processes of IP circulation across platforms, formats, and audience environments (Jenkins, 2006). What remains untheorized, however, is not the fact of circulation but its ecosystem-level mechanics and, in particular, the conditions under which circulation accumulates value for the originating ecosystem rather than concentrating it in dominant intermediaries.

Third, although orchestration is a well-established construct, it has not been extended to platform-mediated cultural industries operating under compound risk and uncertainty. In innovation-network research, orchestration denotes the deliberate actions through which a focal actor mobilizes knowledge, manages appropriability, and maintains network stability without hierarchical authority (Dhanaraj & Parkhe, 2006). In governance research, orchestration denotes an indirect and soft mode of steering in which a governing actor works through intermediaries it cannot fully control (Abbott et al., 2015; Abbott & Snidal, 2010). This literature has developed largely in corporate-network and transnational-regulatory settings. Importantly, it is not the case that existing scholarship neglects the state's role in cultural industries: developmental-state and Korean Wave research documents extensive and active state coordination of Korea's content sector (Jin, 2024; Lee & Lee, 2024). The gap is therefore not the absence of policy from existing accounts, but the absence of a mechanism-level theory of how public actors orchestrate platform-mediated content ecosystems, that is, how they align IP circulation, platform coordination, and network integration under conditions in which calculable risk and irreducible uncertainty compound one another. Addressing this gap is analytically significant because decentralized market coordination alone is often insufficient to sustain ecosystem resilience under volatile digital conditions.

3. Conceptual framework

3.1. Defining content innovation ecosystems

Building on the conceptual gaps identified in the preceding discussion, this study defines the content innovation ecosystem as a distinct form of adaptive innovation system characterized by uncertainty-intensive value creation, platform-mediated coordination, and the centrality of intangible intellectual assets. While existing innovation ecosystem frameworks emphasize interdependencies among actors and complementarities in value creation (Adner, 2017; Jacobides et al., 2018), they have largely been developed in technological and manufacturing contexts where innovation is driven by modular architectures, technological integration, and relatively stable coordination environments. However, digital content industries operate under fundamentally different conditions characterized by symbolic value, recursive circulation, volatile audience attention, and infrastructural dependency on digital platforms (Autio et al., 2018). Extending ecosystem theory to these contexts, content innovation ecosystems are conceptualized here as configurations of interdependent actors, institutions, and artifacts that collectively enable the creation, circulation, coordination, and adaptive amplification of intellectual property within digital cultural industries. This definition shifts the analytical focus from

technological components to intellectual property as the primary unit of ecosystem value and adaptation.

This conceptualization highlights three interrelated dimensions that distinguish content ecosystems from conventional innovation systems. First, value creation is not anchored in discrete production activities or linear innovation trajectories, but in the continuous development, recombination, circulation, and extension of intellectual property across interconnected platforms and markets (Jenkins, 2006; Potts et al., 2008). Innovation therefore emerges through recursive and adaptive processes rather than isolated technological events. Second, ecosystem interactions are increasingly mediated through digital platforms that function not only as technological infrastructures, but also as governance architectures that shape visibility, monetization, discoverability, and audience access across global networks (Gillespie, 2018; Nieborg & Poell, 2018; Parker et al., 2016). Third, innovation arises through distributed and networked interactions among multiple stakeholders, including creators, intermediaries, platforms, policy institutions, and audiences, highlighting the relational, participatory, and continuously evolving nature of value creation within digital ecosystems (Autio & Thomas, 2014; Cunningham & Craig, 2019).

3.2. Structural components of the ecosystem

Content innovation ecosystems are organized around three interdependent structural components. First, intellectual property (IP) represents the primary strategic resource around which content ecosystems are organized. Unlike physical goods or modular technological components, IP embodies symbolic meanings, creative narratives, and intangible cultural assets that can be continuously reproduced, recombined, and extended across multiple formats, platforms, and markets (Caves, 2000; Towse, 2010). As discussed previously, IP is not merely an output of creative activity, but a reusable, scalable, and recursively circulating asset that underpins long-term ecosystem value creation. Under conditions of digital uncertainty and rapidly shifting audience attention, the adaptive capacity of IP becomes particularly important because successful intellectual assets must continuously evolve to maintain relevance, visibility, and monetization across changing platform environments. In this sense, IP functions not only as the central resource of the ecosystem, but also as a stabilizing mechanism that enables recursive innovation and sustained ecosystem engagement over time.

Second, platforms provide the infrastructural architecture through which interactions among ecosystem participants are coordinated and governed. Digital platforms such as streaming services, social media infrastructures, and content-sharing ecosystems facilitate the distribution, visibility, monetization, and circulation of content across global networks (Gawer, 2014; Nieborg & Poell, 2018; Parker et al., 2016). By enabling multi-sided interactions among creators, firms, audiences, advertisers, and intermediaries, platforms coordinate economic activity without relying on traditional hierarchical control. However, platforms are not neutral intermediaries. They increasingly function as governance infrastructures that shape ecosystem participation through algorithms, recommendation systems, data analytics, and monetization architectures (Gillespie, 2018; Plantin et al., 2018). As platform owners accumulate infrastructural power, ecosystem actors become increasingly dependent on opaque and asymmetrical governance environments characterized by changing rules, algorithmic uncertainty, and concentrated control over audience access and visibility (Srnicek, 2017; Zuboff, 2019). Platforms therefore simultaneously enable ecosystem coordination and generate new forms of dependency, vulnerability, and governance complexity.

Third, networks constitute the connective infrastructure linking actors, resources, and capabilities across the ecosystem. Innovation in content industries is inherently collaborative, distributed, and interdependent, involving a wide range of participants who contribute diverse forms of expertise, creativity, and symbolic knowledge (Granovetter,

1985; Powell et al., 1996). Networks facilitate the circulation of information, talent, cultural meanings, and complementary resources, enabling processes such as co-creation, collaborative experimentation, and collective problem-solving. Importantly, networks within content ecosystems are often fluid, project-based, and temporally unstable, reflecting the uncertain and iterative nature of creative production (DeFillippi et al., 2007; Grabher, 2002). Under such conditions, network adaptability and connectivity become critical determinants of ecosystem resilience because they shape how effectively actors can mobilize complementary capabilities and respond to rapidly changing market and technological environments.

These three components, IP, platforms, and networks, constitute the ecosystem's resource and coordination base. They do not, however, align themselves. A fourth element, policy orchestration, operates at a higher order to govern and align them and is therefore treated separately as a governance mechanism in Section 3.4 rather than as a peer structural component. Distinguishing the three first-order components from the higher-order orchestration layer is deliberate: it prevents the conflation of the ecosystem's coordination base with the governance process that acts upon it, and it fixes the level at which each element operates.

3.3. Key generative mechanisms of value creation

Three generative mechanisms drive value creation within content innovation ecosystems, each activating one of the three structural components. The first mechanism, IP circulation, captures the dynamic and recursive process through which intellectual property is continuously reused, recombined, adapted, and extended across multiple formats, platforms, and markets. Unlike traditional innovation models that emphasize the creation of entirely new products or technologies, innovation in content industries frequently emerges through iterative transformations of existing intellectual assets. Practices such as adaptation, serialization, franchising, and transmedia expansion enable IP to evolve over time, generating cumulative value through repeated circulation, reinterpretation, and audience engagement (Caves, 2000; Jenkins, 2006). Under uncertainty-intensive digital environments, the recursive circulation of IP functions not only as a mechanism of value generation, but also as a mechanism of ecosystem resilience by sustaining relevance, scalability, and long-term engagement across volatile markets. Crucially, however, this value-generating dynamic is not automatic. Because circulation depends on who holds the rights to recombine and extend an asset, the outcome of IP circulation is contingent on the allocation of IP ownership. Where originators retain ownership, recursive circulation accumulates value for the ecosystem's productive base; where ownership is transferred to a dominant platform, the same circulation can concentrate downstream value in the platform and reproduce dependency for originators, a contingency examined empirically in Section 5. This conceptualization departs from conventional perspectives that treat IP primarily as a static asset for protection or appropriation (Teece, 1986; Towse, 2010).

The second mechanism, platform coordination, refers to the role of digital platforms in organizing, governing, and stabilizing interactions among ecosystem participants under conditions of distributed and uncertain coordination. Platforms function as infrastructural intermediaries that connect producers, consumers, advertisers, creators, and complementary actors through algorithmic and data-driven systems (Gawer & Cusumano, 2014; Parker et al., 2016; Srnicek, 2017). Through recommendation algorithms, monetization systems, data analytics, and visibility architectures, platforms reduce transaction costs and facilitate large-scale coordination among heterogeneous participants who often lack direct contractual relationships. Platforms also increasingly function as governance infrastructures that define the rules, standards, and conditions under which ecosystem interactions occur (Gillespie, 2018; Tiwana, 2014). Algorithmic visibility influences audience attention, revenue-sharing models shape value appropriation, and platform policies determine access, participation, and discoverability. At the same

time, these governance mechanisms generate new forms of dependency, asymmetry, and ecosystem vulnerability because dominant platforms control critical infrastructures and data flows (Plantin et al., 2018; Zuboff, 2019). Platform coordination therefore simultaneously enables ecosystem integration and reproduces uncertainty-intensive governance conditions within digital innovation environments.

The third mechanism, network integration, emphasizes the role of relational connectivity, collaborative interaction, and ecosystem adaptability in enabling innovation under volatile conditions. Content innovation ecosystems depend heavily on the ability of diverse actors, including creators, producers, distributors, audiences, and institutional actors, to combine complementary capabilities, symbolic resources, and creative knowledge (Granovetter, 1985; Powell et al., 1996). Network integration facilitates these processes by enabling knowledge exchange, collaborative experimentation, resource mobilization, and collective problem-solving across organizational and institutional boundaries. Importantly, these networks are often fluid, project-based, and temporally unstable, reflecting the iterative and uncertain nature of digital cultural production (DeFillippi et al., 2007; Grabher, 2002). Under such conditions, ecosystem resilience depends significantly on the degree of network adaptability, connectivity, and diversity. Network integration therefore functions as the relational mechanism through which ecosystems sustain adaptive coordination and regenerate innovation capacity.

Although platform coordination and network integration both coordinate heterogeneous actors, they are analytically distinct along three boundary conditions, which establish their discriminant validity. First, they differ in the locus of coordination: platform coordination is vertical and infrastructural, channeled through a dominant intermediary that owns the coordinating infrastructure, whereas network integration is horizontal and relational, negotiated directly among peer actors. Second, they differ in the mode of governance: platform coordination operates through algorithmic and contractual rules that a platform sets and can alter unilaterally, whereas network integration operates through reciprocal, project-based agreements among participants. Third, they differ in the direction of dependency: platform coordination concentrates infrastructural control and tends to render participants dependent on a gatekeeper, whereas network integration distributes coordination capacity and tends to preserve participant autonomy. Observable indicators follow accordingly. Platform coordination is evidenced by algorithmic recommendation, visibility and ranking control, platform-set revenue-sharing terms, and commissioning by dominant OTT operators, whereas network integration is evidenced by inter-firm co-productions, creator collaborations, cross-sector IP partnerships, and talent and agency networks that are not gatekept by a controlling platform.

3.4. Governance and policy orchestration

Governance in content innovation ecosystems operates through orchestration, understood here as a higher-order mechanism that acts upon the three generative mechanisms rather than as a fourth mechanism operating alongside them. A central dimension of orchestration involves the development of coordination mechanisms capable of connecting fragmented ecosystem components under conditions of complexity and volatility. These mechanisms include financial instruments, regulatory arrangements, intellectual property systems, standard-setting processes, intermediary organizations, and institutional platforms that facilitate collaboration, experimentation, and information exchange. In digital content industries, where production is highly project-based, platform-mediated, and dependent on heterogeneous actors, coordination challenges are especially pronounced. Ecosystem participants often face uncertainties related to platform governance, audience visibility, revenue distribution, technological change, and access to infrastructural resources. Under these conditions, orchestration plays a role in reducing coordination failures, lowering transaction costs, enabling resource accessibility, and facilitating

collaborative adaptation across ecosystem actors. Importantly, coordination in this context is not hierarchical but relational, distributed, and continuously recalibrated through indirect governance mechanisms such as incentives, infrastructures, standards, and policy signals (Gawer, 2021; Jacobides et al., 2018).

Crucially, governance in digital content ecosystems operates under conditions of continuous uncertainty, rapid technological transformation, and evolving platform environments where outcomes cannot be fully predicted and static regulatory approaches quickly become inadequate. This necessitates adaptive and iterative forms of governance characterized by experimentation, feedback, institutional learning, and ongoing recalibration (Ansell & Boin, 2019; Mazzucato, 2018). Ecosystem orchestration is therefore not a one-time policy intervention, but a recursive and co-evolving governance process that develops alongside ecosystem transformation itself. The ability of institutional actors to monitor ecosystem dynamics, respond flexibly to emerging disruptions, learn from policy outcomes, and continuously redesign governance instruments becomes essential for sustaining coordination, resilience, and long-term innovation capacity within uncertainty-intensive digital environments.

This orchestration performs two analytically distinct functions that correspond to Knight's (1921) two categories. Against calculable risk, that is, the diversifiable and probabilistically tractable exposures of production-cost volatility, project financing, and portfolio-level commercial risk, orchestration works primarily through policy-finance instruments such as funds, guarantees, and co-investment schemes that pool and redistribute risk across the ecosystem. Against irreducible uncertainty, that is, the non-probabilizable exposures of unpredictable audience demand, generative AI disruption, platform restructuring, and contested IP appropriation, orchestration works primarily through its adaptive and iterative character: experimentation, monitoring, institutional learning, and continuous recalibration of instruments as conditions change. Conceptualized in the terms of innovation-network and governance research, policy orchestration extends the orchestration construct, understood as the deliberate mobilization of resources and stabilization of relations by a focal actor that lacks hierarchical control (Abbott & Snidal, 2010; Dhanaraj & Parkhe, 2006), from corporate networks and transnational regulation to platform-mediated cultural ecosystems, where the orchestrator is a public actor and the object of alignment is the interaction among IP circulation, platform coordination, and network integration. Because the orchestrator works through intermediaries and incentives it does not fully control, orchestration is best understood as a set of attempts to align and stabilize the ecosystem, whose effects are contingent rather than guaranteed.

4. Conceptual approach and illustrative case

4.1. Research design: A conceptual article with an illustrative case

This study is a conceptual, theory-building contribution rather than an empirical test or an inductive coding study. Its aim is to develop a framework that reconceptualizes ecosystem structures, generative mechanisms, and governance processes in platform-mediated cultural industries under conditions of compound risk and uncertainty. Following Jaakkola's (2020) typology of conceptual articles, the study combines theory synthesis, integrating innovation ecosystem theory, platform studies, and cultural economics into a single framework, with theory adaptation, extending the orchestration construct to a new empirical domain (Cornelissen, 2017; Jaakkola, 2020). Consistent with this genre, the framework is developed first, through conceptual reasoning across literatures, and is then illustrated, rather than discovered or tested, through an interpretive reading of a single revelatory case. This sequencing is a feature of the conceptual-article design, not a methodological inconsistency: the case serves to demonstrate the framework's plausibility and analytical purchase, not to generate it inductively or to confirm it statistically.

4.2. The illustrative case and its evidentiary base

The illustrative case focuses on the Korean content industry during the period from 2023 to the first half of 2026. This period represents a critical phase of political-economic and institutional restructuring associated with rapid platformization, global OTT expansion, intensifying competition over intellectual property (IP), and major transformations in Korea's digital content governance system. During this period, substantial policy and industrial changes occurred across ecosystem governance, content regulation, digital infrastructure policy, taxation, research and development investment, public-private partnerships, and global content expansion strategies. These developments accelerated ecosystem restructuring and generated heightened uncertainty regarding platform governance, value distribution, industrial coordination, and market sustainability.

Korea was selected as a revelatory and analytically rich case because it constitutes one of the world's most advanced digital content ecosystems, characterized by strong platform integration, intensive IP circulation, active state intervention, and globally networked cultural production (Eisenhardt & Graebner, 2007; Siggelkow, 2007). Simultaneously, the Korean ecosystem exhibits tensions associated with infrastructural dependency, platform asymmetry, ecosystem fragmentation, and uncertainty-intensive innovation, making it theoretically appropriate for illustrating the proposed mechanisms and, in particular, their contingencies.

The evidentiary materials for the illustration were assembled through purposive and theoretically informed sampling (Patton, 2014). The sampling process prioritized documents that were repeatedly referenced and utilized by policymakers, public institutions, industry organizations, investment actors, strategic planning agencies, and ecosystem governance stakeholders in the formulation of digital content policies, industrial strategies, investment frameworks, and ecosystem performance management systems. To enhance analytical relevance and ecosystem representativeness, document selection was additionally refined through expert consultation with researchers, policy specialists, and practitioners specializing in digital content industries, platform governance, and innovation ecosystem strategy.

The final corpus consisted of 80 documents published between 2023 and 2026, including government policy plans, institutional strategy reports, public agency publications, platform industry analyses, market intelligence reports, securities and venture capital analyses, consulting reports, and peer-reviewed academic studies. Specifically, the corpus included 30 government and public policy documents, 25 industry and platform reports, 10 investment and market analyses, and 15 academic and theoretical studies (see Table 1). Major institutional sources included the Ministry of Culture, Sports and Tourism (MCST), Korea Creative Content Agency (KOCCA), Korea Communications Commission (KCC), Korea Information Society Development Institute (KISDI), Korean Film Council (KOFIC), major platform-related corporate analyses, and international consulting and market intelligence organizations. These materials constitute the evidentiary base for illustrating the framework rather than a dataset from which the framework was induced.

4.3. Organizing the illustrative evidence

Because the framework was specified before the case, the analytical task was not to induce categories from the data but to organize the evidentiary materials against the framework and to assess their fit. The materials were read thematically and mapped onto the four framework elements, namely the three generative mechanisms and the higher-order orchestration layer, with attention to whether the Korean evidence corroborated, qualified, or contradicted each element. Where the evidence pointed against the framework's expectations, for example the persistence of buy-out contracts and the deepening of polarization, these tensions were retained rather than smoothed over, and were used to

Table 1
Composition of the evidentiary corpus (2023–2026).

Source Category	Representative Institutions/Outlets	Number of Documents	Primary Analytical Focus
Government and public policy documents	Ministry of Culture, Sports and Tourism (MCST); Korea Creative Content Agency (KOCCA); Korea Communications Commission (KCC); Korea Copyright Commission; Korean Film Council (KOFIC)	30	Policy orchestration; IP and AI regulation; ecosystem governance
Industry and platform reports	Korea Information Society Development Institute (KISDI); KOCCA industry trend reports; Korea Institute for Industrial Economics and Trade (KIET); platform-related industry analyses	25	Platformization; OTT dynamics; production cost structures; algorithmic governance
Investment and market analyses	Securities and venture capital research notes; market intelligence reports; consulting and industry forecast publications	10	Capital flows; IP investment; market concentration; risk distribution
Peer-reviewed academic studies	Journals including Telecommunications Policy; Media, Culture & Society; Convergence; International Journal of Communication; R&D Management	15	IP circulation; transmedia and webtoon ecosystems; ecosystem theory; uncertainty management
Total		80	

Note. Document counts represent purposively sampled materials repeatedly referenced across policy, industry, and academic discourse on the Korean digital content ecosystem (2023–2026). Some documents are cross-classified and counted under their primary analytical orientation.

sharpen the framework's contingency claims rather than to confirm a uniform stabilizing effect.

The result of this procedure is summarized in Table 2 as a framework-evidence correspondence, which records, for each mechanism, its conceptual definition and the illustrative evidence, both confirming and disconfirming, drawn from the Korean case. Table 2 should therefore be read as a correspondence between concepts and illustrative evidence, not as the output of an inductive coding procedure. Presenting the correspondence in this form makes the basis of the illustration transparent and allows readers to judge the fit between each framework element and the evidence adduced for it.

4.4. Interpretive transparency and rival interpretations

Because the contribution is conceptual and illustrative, its credibility rests on interpretive transparency and analytical plausibility rather than on statistical inference (Siggelkow, 2007; Yin, 2018). Two practices supported this. First, rival interpretations and disconfirming evidence were deliberately retained. Platformization, for instance, was read both as a mechanism of ecosystem coordination and as a source of infrastructural asymmetry and dependency, and the prevalence of buy-out contracts was treated as a genuine counter-tendency to the framework's stabilizing logic rather than as noise. Second, the study adopts analytical rather than statistical generalization. The Korean case is used not as a representative sample but as a revelatory setting through which the framework's mechanisms and their contingencies become analytically visible and, in principle, transferable to other platform-mediated, uncertainty-intensive content ecosystems.

Table 2
Framework-evidence correspondence.

Ecosystem Mechanism	Conceptual Definition	Illustrative Evidence from the Korean Case (2023–2026)
IP circulation (generative)	Recursive reuse, recombination, and transmedia extension of IP across formats and markets; value is contingent on the allocation of IP ownership.	Confirming: webtoon-to-drama, film, and game pipelines (Naver Webtoon, Kakao Page); increasingly cross-sectoral IP transactions (KOCCA, 2025); portfolio-oriented IP strategies. Disconfirming: buy-out contracts transferring global rights to OTT platforms, leaving originators without residual rights (“premium subcontracting”).
Platform coordination (generative)	Vertical, infrastructural coordination governed by dominant platform intermediaries through algorithmic visibility, monetization, and access control.	Scale of OTT investment (Netflix, USD 2.5 billion, 2023–2026); concentration of monthly active users (Netflix, 14.2 million); production, distribution, and monetization increasingly governed through algorithmic systems (KISDI, 2025).
Network integration (generative)	Horizontal, relational coordination among peer actors through inter-firm co-productions, creator collaborations, and cross-sector partnerships not gatekept by a controlling platform.	Project-based partnerships among webtoon studios, production companies, talent agencies, and game developers assembling adaptation pipelines; recomposable cross-sector configurations reassembled as projects and market conditions change.
Policy orchestration (higher-order)	Indirect, adaptive governance that aligns the three generative mechanisms; pools calculable risk through policy finance and absorbs irreducible uncertainty through iterative recalibration.	Global League Fund (KRW 100 to 150 billion) designed to preserve domestic IP ownership; Future Strategy Fund; K-Content Strategic Fund (KRW 1.02 trillion); AI Basic Act (2025) and Korea Copyright Commission guidelines; Video Content Compensation Consultative Body (2024). Limits: uneven legislative progress and persistent buy-out dependency.

Note. The table records, for each framework element, its conceptual definition and the illustrative evidence, both confirming and disconfirming, drawn from the Korean case. It represents a correspondence between concepts and illustrative evidence, not the output of an inductive coding procedure.

5. Empirical illustration: The Korean content industry

5.1. Ecosystem restructuring under platformization and uncertainty

Recent policy documents, institutional reports, and industry analyses published between 2023 and 2026 consistently characterized the Korean content industry as undergoing a period of accelerated ecosystem restructuring driven by platformization, AI-enabled digital transformation, intensifying global competition, and increasing uncertainty regarding governance, monetization, and value distribution. Rather than describing the industry simply as a successful export-oriented cultural sector, many of the analyzed materials emphasized the growing instability and complexity of the ecosystem itself, particularly in relation to platform dependency, algorithmic governance, concentration of value capture, and fragmentation across production networks. Reports issued by the Ministry of Culture, Sports and Tourism (MCST), the Korea Creative Content Agency (KOCCA), and multiple market intelligence organizations repeatedly highlighted the need for

adaptive ecosystem coordination and institutional restructuring capable of responding to rapidly changing platform environments and transnational market volatility.

The scale and trajectory of these transformations are visible in recent industry statistics. According to a recent government survey (Ministry of Culture, Sports & Tourism, 2026), total revenue in the Korean content industry reached approximately KRW 157.4 trillion (about USD 113 billion) in 2024, a 2.1 percent increase over the previous year, while content exports grew to approximately USD 14.07 billion. The industry employed about 688,000 workers across roughly 121,000 enterprises spanning eleven sectors, including broadcasting and video, knowledge-information services, publishing, games, music, animation, advertising, and character and licensing. Among these, broadcasting and video (KRW 24.99 trillion), knowledge-information (KRW 24.70 trillion), publishing (KRW 24.22 trillion), and games (KRW 23.85 trillion) accounted for the largest shares, indicating that growth has been broadly distributed rather than concentrated in a single genre. Industry reports during this period repeatedly emphasized that this aggregate expansion masked deepening structural tensions associated with platform dependency, IP appropriation, and rising production costs (Korea Development Institute, 2024; Korea Creative Content Agency, 2024).

At the same time, the analyzed materials consistently highlighted growing concerns regarding ecosystem fragmentation and uneven coordination. Despite rapid global expansion, the Korean content ecosystem remains characterized by polarization between dominant platform-integrated actors and smaller independent producers, increasing production cost pressures, unstable labor conditions, and asymmetrical bargaining relationships surrounding IP ownership and monetization. Multiple policy and institutional reports described these tensions as systemic rather than temporary, emphasizing that market-driven coordination alone was insufficient to maintain ecosystem sustainability under conditions of accelerating technological and market uncertainty. These observations reinforced growing institutional recognition of the need for adaptive governance frameworks intended to stabilize ecosystem interactions, reduce coordination failures, and mitigate structural asymmetries across platform-centered markets.

5.2. Adaptive governance and policy orchestration under ecosystem uncertainty

The Korean institutional landscape has increasingly shifted toward adaptive policy orchestration. Several recent policy initiatives illustrate this logic in concrete form. The Korean government's 2025 plan to establish a KRW 100 billion (about USD 72 million) Global League Fund, subsequently scaled up to KRW 150 billion in 2026, was designed specifically to channel foreign capital into Korean content projects while preserving domestic IP ownership (Ministry of Culture, Sports & Tourism, 2025a, 2025b). In parallel, the Future Strategy Fund (KRW 100 billion in 2026) was created to remove existing restrictions limiting investment to broadcasting and video, thereby enabling cross-genre, cross-platform IP development. These initiatives, together with the KRW 1.02 trillion K-Content Strategic Fund (over five years from 2024) and an expanded content industry guarantee program of KRW 220 billion, indicate that policy finance is increasingly mobilized not as compensatory support but as an active instrument for shaping the direction of IP development, platform integration, and ecosystem coordination (Ministry of Culture, Sports & Tourism, 2025a).

In Knightian terms, these instruments target primarily calculable risk. By pooling project-level financing and completion risk across many productions, and by diversifying exposure through cross-genre and cross-platform mandates, funds and guarantees redistribute exposures that are, in principle, probabilistically tractable. Notably, the explicit design of the Global League Fund to preserve domestic IP ownership indicates that policy finance is being used not only to pool risk but also to shape the ownership contingency discussed below. These instruments do not, and cannot, eliminate the irreducible uncertainties of audience

reception or platform restructuring; those exposures are addressed, if at all, through the adaptive and iterative recalibration of policy discussed throughout this section.

One of the most significant dimensions of this institutional transformation concerns policy finance. Recent policy and industry reports consistently identified structural financing asymmetries within the content ecosystem, particularly regarding independent creators, small production firms, and early-stage IP development. Because content production is characterized by uncertain returns, project-based risk, and intangible asset dependency, private investment often remains concentrated around already successful IP portfolios and platform-integrated firms. In response, Korean policy institutions increasingly expanded targeted funding programs, public-private investment schemes, content-focused venture financing, and strategic ecosystem funds intended to reduce market uncertainty and stabilize innovation capacity. Importantly, the analyzed materials revealed that policy finance was no longer framed merely as compensatory support for market failure, but as a strategic instrument for ecosystem activation and selective coordination. Financial interventions increasingly shaped which forms of content production, technological convergence, and platform integration received institutional prioritization, thereby seeking to influence the direction of IP development and ecosystem evolution itself (Kattel & Mazzucato, 2018; Mazzucato, 2018).

Intellectual property policy emerged as another central area of adaptive governance tension. Multiple institutional and industry reports emphasized increasing uncertainty surrounding IP ownership, platform-based monetization, AI-generated content, transmedia circulation, and cross-platform licensing structures. These developments intensified tensions between protection and openness within the ecosystem. On the one hand, stronger IP protection remained necessary to incentivize investment and secure value appropriation in highly competitive global markets. On the other hand, excessive exclusivity risked constraining recombination, collaborative production, and ecosystem-level innovation. The analyzed policy discussions repeatedly framed IP governance not simply as a legal issue of ownership protection, but as a broader coordination challenge involving circulation, scalability, interoperability, and adaptive ecosystem sustainability. In this regard, IP policy increasingly shaped the conditions under which intellectual assets circulated across platforms, networks, and transnational markets, thereby directly influencing the ecosystem mechanism of IP circulation identified in this study.

5.3. Platformization, algorithmic governance, and ecosystem dependency

The empirical scale of platform dependency in the Korean ecosystem is illustrated by the rapid expansion of global OTT investment. Netflix announced plans to invest USD 2.5 billion in Korean content between 2023 and 2026, and in its 2025 Korean content lineup announced 21 scripted series, 8 unscripted series, and 8 original films produced in Korea, reflecting an unprecedented scale of foreign investment in the Korean content ecosystem (Korea Creative Content Agency, 2024; Kim, W., 2025). Per-episode budgets have escalated correspondingly: while early Korean originals such as *Kingdom* (2019) were produced at approximately USD 1.7 million per episode, the second season of *Squid Game* (2024) reportedly reached close to USD 9.8 million per episode. As of January 2025, monthly active users for the leading OTT services in Korea included Netflix (14.2 million), Coupang Play (7.6 million), Tving (6.3 million), Wavve (2.7 million), and Disney+ (2.4 million), illustrating both the dominance of foreign platforms and the structural difficulty domestic services face in matching their scale (Korea Information Society Development Institute, 2025). These figures repeatedly emerged in analyzed industry reports as evidence that platform coordination has become inseparable from ecosystem governance itself.

The analyzed materials consistently revealed that digital platforms increasingly function as infrastructural coordinators governing interactions among creators, firms, audiences, advertisers, and

intermediaries across the ecosystem. Unlike traditional broadcasting systems characterized by relatively stable and sequential value chains, platform-mediated environments operate through real-time, data-driven, and algorithmically organized interactions that continuously restructure relationships among ecosystem actors (Gawer & Cusumano, 2014; Parker et al., 2016). Multiple industry reports emphasized that production, distribution, audience engagement, and monetization processes were becoming increasingly integrated within platform-centered infrastructures where visibility, discoverability, and market access are governed through algorithmic systems rather than through conventional institutional coordination mechanisms. This vertical, gatekept form of coordination is analytically distinct from the horizontal, peer-based coordination captured by network integration, discussed below in relation to webtoon-to-drama collaborations.

5.4. Recursive IP circulation and the ownership contingency

Empirical patterns within the Korean webtoon and transmedia segments illustrate the centrality of recursive IP circulation. Webtoon platforms such as Naver Webtoon and Kakao Page operate as upstream IP origination sites whose works are systematically adapted into K-dramas, films, mobile games, and merchandise, creating layered revenue streams from a single intellectual asset (Jin, 2023; Lee & Lee, 2024; Pyo et al., 2019). Industry analyses noted that webtoon-derived adaptations now feed content pipelines spanning Netflix, Sony, and other global studios, with Naver's 2021 acquisition of Wattpad embedding the Korean webtoon model within transnational distribution networks. According to the Korea Creative Content Agency's 2024 IP Transaction Status Survey (Korea Creative Content Agency, 2025), content IP transactions across broadcasting, music, comics and webtoons, animation, character and licensing, and games have become increasingly cross-sectoral, reflecting a shift toward portfolio-oriented IP strategies. These patterns provide consistent evidence that value in the Korean content ecosystem is generated not through discrete content production, but through systematized and recursive IP circulation across formats and markets. Notably, these adaptation pipelines are frequently assembled through horizontal, project-based partnerships among webtoon studios, production companies, talent agencies, and game developers, illustrating network integration as a mechanism distinct from platform-gatekept coordination.

A particularly important pattern concerns the recursive nature of IP circulation. The analyzed reports repeatedly highlighted that successful Korean content IP increasingly moves across multiple interconnected formats, including streaming series, films, webtoons, gaming, music, merchandise, fan communities, and immersive digital environments, through ongoing processes of recombination and transmedia expansion. Importantly, these circulation processes are not linear but cumulative and path-dependent. Each new adaptation or platform extension not only generates additional value but also strengthens the symbolic visibility, audience recognition, and commercial viability of the original intellectual asset itself. This recursive dynamic reflects broader theories of cumulative innovation and dynamic capability formation, in which ecosystem value emerges through continuous recombination and redeployment of existing intangible assets rather than through isolated production events (Nelson & Winter, 1982; Teece, 2007).

At the same time, the analyzed materials repeatedly identified IP appropriation as one of the most contested sources of structural uncertainty within the ecosystem. Industry analyses and academic studies pointed to the prevalence of “buy-out” contracts under which global OTT platforms cover full production costs in exchange for exclusive global IP ownership, leaving Korean producers without residual rights to derivative works, merchandise, sequels, or transmedia adaptations (Korea Creative Content Agency, 2025; Kim, T., 2022; Kim, W., 2025; Lee, 2024). Even when Korean originals achieve global success, downstream value accrues primarily to platforms rather than to originating production companies. In response, the Video Content Compensation

Consultative Body was launched in 2024 and legislative discussions on an additional compensation right for audiovisual content remain ongoing, although progress has been uneven.

This pattern makes the ownership contingency concrete. The recursive circulation that generates cumulative value in the webtoon-to-drama pipelines, where Korean firms typically retain origination rights, is the same mechanism that, under buy-out contracts transferring global rights to the commissioning platform, concentrates downstream value in the platform and leaves originators without residual claims. IP circulation is therefore not intrinsically value-generating for the originating ecosystem; whether it accumulates value domestically or reproduces dependency depends on where ownership is allocated. The framework accordingly treats IP ownership allocation as the decisive contingency governing whether IP circulation stabilizes or destabilizes the originating ecosystem, and it reads the contrast between retained-rights webtoon pipelines and rights-transferring buy-out productions as the empirical expression of that contingency.

5.5. Ecosystem fragmentation and structural fragility under platformization

The combination of production cost inflation and platform concentration has visibly reshaped the volume and structure of Korean content production. South Korea produced approximately 141 dramas in 2022, but this figure has since contracted sharply as production cost escalation and distribution bottlenecks have intensified (Korean Film Council, 2025; Kim, W., 2025). Theatrical revenue rebounded from approximately KRW 510 billion in 2020 (a pandemic low) to approximately KRW 1.19 trillion in 2024, while remaining well below pre-pandemic levels, with 2024 marking a 5.3 percent decline from the previous year (Korean Film Council, 2025). Korean-language films accounted for KRW 691 billion (about USD 500 million) of the 2024 total, reflecting partial recovery to roughly 74 percent of the 2017 to 2019 pre-pandemic average, while OTT consumption continued to expand and increasingly substitute for theatrical viewing during the same period (Korea Information Society Development Institute, 2025; Korean Film Council, 2025). Industry reports increasingly described the resulting configuration as a polarized ecosystem in which a small number of platform-integrated production companies command rising budgets while smaller and independent producers face progressive marginalization. This dynamic was repeatedly characterized as a structural condition of “premium subcontracting,” in which Korean producers create high-value content without retaining the IP that drives long-term ecosystem value (Korea Creative Content Agency, 2025; Kim, T., 2022; Lee & Lee, 2024). These outcomes, contraction in drama volume, deepening polarization, and value leakage under buy-out contracts, are read here not as anomalies but as the negative-outcome branch of the framework, the branch that prevails when ownership is transferred and orchestration does not offset the resulting asymmetries.

6. Toward a conceptual model of content innovation ecosystems

6.1. Model synthesis

The proposed model integrates three generative layers, coordinated by a higher-order orchestration layer. First, the model identifies an IP-centered layer as the foundational site of value creation. Content industries increasingly organize value creation around intellectual property as the central unit of accumulation, circulation, and adaptive scalability (Caves, 2000; Towse, 2010). The empirical illustration repeatedly showed that Korean content firms, platforms, and institutional actors increasingly prioritized expandable IP portfolios capable of recursive recombination and transmedia extension across streaming platforms, gaming, webtoons, music, merchandise, immersive media, and global digital markets. Under conditions of uncertainty-intensive competition and volatile audience demand, IP functions not merely as

a static legal asset, but as a dynamic mechanism of adaptive value stabilization that can enable ecosystem actors to sustain engagement, mitigate market uncertainty, and continuously regenerate value through recursive circulation (Jenkins, 2006; Teece, 2007). As shown in the Korean case, mechanisms such as serialization, adaptation, franchising, and transmedia storytelling can generate cumulative and path-dependent ecosystem value over time (Nelson & Winter, 1982), but only where ownership of the circulating asset is retained by the originating ecosystem; where it is transferred, the same layer becomes a conduit for value leakage.

Second, the model emphasizes a platform-centered layer as the primary organizing infrastructure of the ecosystem. The empirical illustration consistently revealed that digital platforms increasingly function as infrastructural governors rather than passive intermediaries, organizing interactions among creators, firms, audiences, advertisers, and institutional actors through algorithmic coordination systems. Through recommendation algorithms, behavioral data analytics, engagement metrics, and multi-sided market architectures, platforms structure visibility, monetization opportunities, and access to audiences across the ecosystem (Gawer & Cusumano, 2014; Parker et al., 2016; Srnicek, 2017). In this regard, platforms operate simultaneously as mechanisms of scalability and sources of structural dependency. While platform integration enables global circulation and accelerated ecosystem expansion, it also intensifies asymmetrical governance relationships by concentrating infrastructural control, audience data, and monetization systems within dominant platform actors (Kenney & Zysman, 2016; Van Dijck et al., 2018). The platform-centered layer therefore constitutes both the connective backbone and a major source of instability within the ecosystem, continuously reshaping how value is distributed, coordinated, and appropriated under volatile digital conditions.

Third, the model specifies a network-centered layer of horizontal, relational coordination among peer actors. Distinct from platform coordination, which runs through and is governed by dominant intermediaries, network integration operates through inter-firm co-productions, creator collaborations, cross-sector IP partnerships, and project-based teams that are negotiated directly among participants (Granovetter, 1985; Powell et al., 1996). In the Korean case, these relational networks link webtoon studios, drama producers, talent agencies, game developers, and music labels into recomposable configurations that can be reassembled as projects and market conditions change. This adaptability is the ecosystem's principal relational resource for absorbing uncertainty that neither IP portfolios nor platform infrastructures can absorb on their own.

Fourth, the model introduces policy-centered orchestration as a higher-order governance layer that seeks to align ecosystem components and to stabilize structurally fragmented interactions. As shown throughout the illustration, content innovation ecosystems are characterized by persistent uncertainty, uneven adaptive capacity, recursive volatility, and interdependence that cannot be fully resolved through decentralized market coordination alone (Kattel & Mazzucato, 2018; Mazzucato, 2018). The analyzed policy materials repeatedly emphasized growing institutional concern regarding ecosystem fragmentation, creator dependency, platform concentration, governance asymmetry, and long-term sustainability under rapidly changing digital conditions. In response, policy increasingly operates not merely as industrial support or market correction, but as a system-level orchestration process intended to facilitate adaptive coordination across heterogeneous ecosystem actors. Financial support systems, research and development programs, platform governance initiatives, IP regulation frameworks, and public-private coordination mechanisms are collectively designed to reduce uncertainty, align incentives, mitigate structural asymmetries, and strengthen ecosystem resilience under conditions of accelerated technological transformation (Rodrik, 2004). Consistent with the orchestration construct, however, these are attempts that work through intermediaries and incentives the state does not fully control (Abbott & Snidal, 2010; Dhanaraj & Parkhe, 2006); their effects are contingent,

and the Korean evidence shows both partial successes, such as the ownership-preserving design of recent funds, and persistent failures, such as continued buy-out dependency.

6.2. From fragmentation to orchestration

While the preceding section presents the structural configuration of content innovation ecosystems, it is equally important to understand how such ecosystems evolve under conditions of uncertainty-intensive digital transformation. Innovation ecosystems are not static or equilibrium-based configurations, but adaptive and continuously reconfigured systems shaped by recursive interactions, technological volatility, institutional adjustment, and asymmetrical coordination dynamics (Adner, 2017; Autio & Thomas, 2014). The illustration of the Korean content industry repeatedly revealed that ecosystems do not emerge as fully integrated structures, but evolve unevenly through processes of fragmentation, coordination, adaptive stabilization, and renewed disruption. Rather than representing a linear trajectory toward stable integration, ecosystem evolution in digital cultural industries is characterized by ongoing tension among scalability, dependency, governance instability, and structural asymmetry. To capture this dynamic, this study conceptualizes ecosystem evolution as a processual transition from fragmentation toward coordination and orchestration, thereby introducing a temporal and uncertainty-sensitive dimension to existing ecosystem theory. Importantly, this transition is neither automatic nor irreversible: as the Korean evidence shows, orchestration attempts can be offset by platform concentration and ownership transfer, returning the ecosystem to fragmentation through the destabilizing feedback loops represented in Fig. 1.

At the initial stage, the ecosystem is characterized by fragmentation, where actors, resources, platforms, and institutional arrangements remain weakly connected and structurally misaligned. Fragmented ecosystems are marked by coordination failures, resource asymmetries, uneven adaptive capacity, and limited complementarities among participants (Adner, 2017; Jacobides et al., 2018). The illustration consistently revealed that fragmentation within the Korean content ecosystem emerged through unequal access to platforms, asymmetrical IP ownership structures, disparities in technological capabilities, and uneven integration into platform-centered value networks. Multiple institutional and policy reports further emphasized that smaller creators and independent firms often faced structural disadvantages in accessing visibility, financing, data infrastructures, and downstream monetization opportunities within increasingly concentrated digital markets. Under these conditions, value creation remains unstable and uneven, relying on isolated successes rather than on sustainable ecosystem complementarities. From a mechanism perspective, fragmentation reflects the underdevelopment, disconnection, or asymmetrical operation of core ecosystem mechanisms: IP circulation constrained or captured by dominant intermediaries, platform coordination unevenly distributed, and network integration structurally incomplete. As a result, ecosystem resources and capabilities remain underutilized, limiting adaptive resilience and long-term innovation capacity.

6.3. Interactions among IP, platforms, and policy

Three triadic interactions structure the governance dynamics of content innovation ecosystems. The first key interaction concerns the relationship between IP and platforms, which together constitute the central site of value creation, circulation, and amplification within the ecosystem. IP provides the symbolic and narrative assets that attract and retain audiences, while platforms enable scalable distribution, algorithmic visibility, audience engagement, and monetization across transnational digital markets (Nieborg & Poell, 2018; Parker et al., 2016). However, the illustration repeatedly revealed that this relationship operates simultaneously as a mechanism of enablement and of dependency. While platforms provide essential infrastructures for global

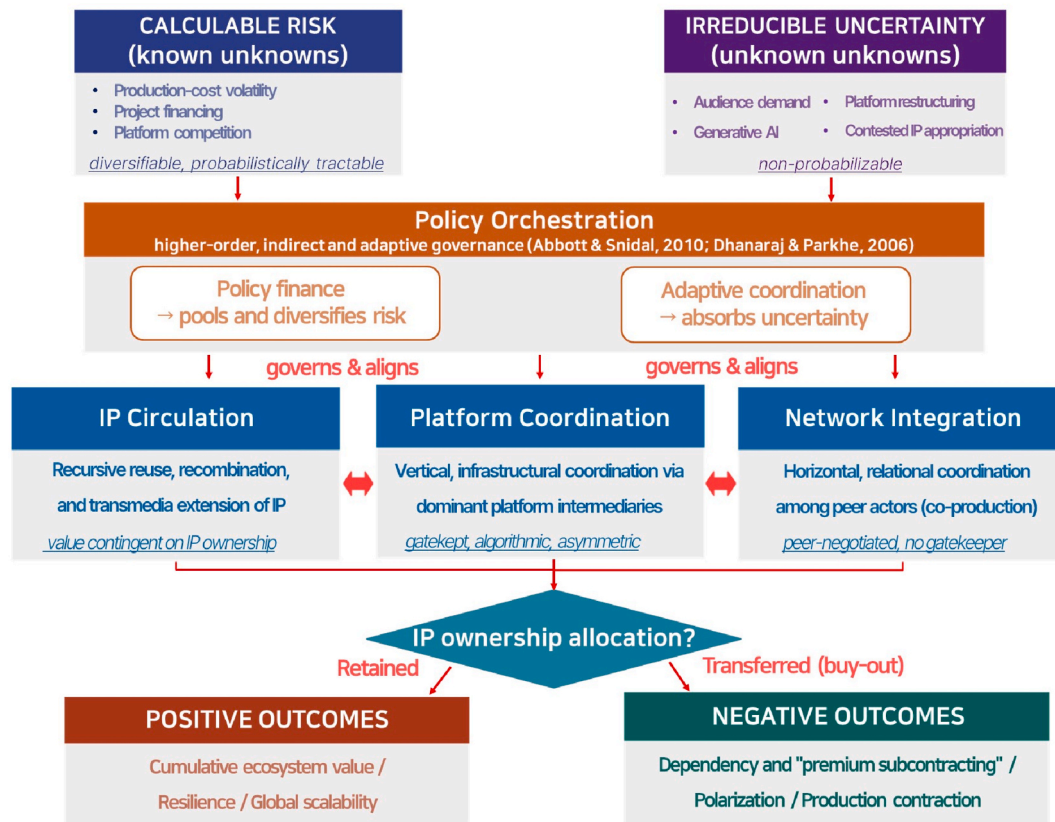


Fig. 1. Conceptual framework of digital content innovation ecosystems under risk and uncertainty. The framework maps Knight's (1921) calculable risk and irreducible uncertainty onto the two functions of policy orchestration (risk pooling through policy finance and uncertainty absorption through adaptive coordination), specifies three generative mechanisms coordinated by a higher-order orchestration layer, and represents both positive outcomes (cumulative value, resilience, global scalability) and negative outcomes (dependency and premium subcontracting, polarization, production contraction), with the IP ownership allocation shown as the contingency gate that determines which outcome prevails. Red arrows trace the flow from Knightian conditions through orchestration and the three generative mechanisms to the IP ownership contingency gate, which routes the ecosystem toward positive or negative outcomes depending on whether IP ownership is retained or transferred. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

scalability and accelerated IP circulation, they also exert significant influence over which forms of IP gain visibility, recommendation priority, investment support, and monetization opportunities through algorithmic curation systems, behavioral data analytics, and engagement-based ranking mechanisms (Srnicek, 2017; Van Dijck et al., 2018). As a result, the IP-platform relationship continuously reproduces asymmetrical governance structures in which value creation becomes increasingly dependent on platform-controlled infrastructures and algorithmically mediated systems of visibility. This is precisely where the ownership contingency operates: when platform commissioning is accompanied by transfer of IP ownership, the IP-platform interaction shifts from enablement toward dependency, as the buy-out evidence demonstrates.

The second interaction involves the relationship between IP and policy, which shapes the institutional conditions under which intellectual assets are created, circulated, protected, and monetized. The illustration repeatedly revealed that policy frameworks increasingly function not merely as legal protections for ownership rights, but as adaptive governance mechanisms shaping the lifecycle, scalability, and circulation logic of intellectual assets across platform-mediated ecosystems. Institutional reports and policy discussions consistently emphasized growing uncertainty regarding AI-generated content, transmedia licensing structures, downstream monetization rights, and cross-platform value appropriation under rapidly evolving digital conditions. These tensions reflected a broader governance dilemma between exclusivity and openness. While stronger IP protection may encourage investment and secure value appropriation, excessive exclusivity may

constrain recombination, interoperability, collaborative production, and ecosystem-level innovation (Gans et al., 2002; Lessig, 2008). Consequently, the IP-policy interaction increasingly operates as a recursive governance mechanism that continuously renegotiates the balance among protection, circulation, accessibility, and adaptive ecosystem sustainability under conditions of technological convergence and market volatility.

The third interaction concerns the relationship between platforms and policy, which together define the evolving governance architecture of digitally transforming ecosystems. As platforms increasingly function as infrastructural governors organizing market access, audience visibility, monetization structures, and ecosystem coordination, issues of accountability, regulation, concentration, and adaptive governance become progressively more significant (Kenney & Zysman, 2016; Van Dijck et al., 2018). The illustration repeatedly demonstrated that policy institutions increasingly recognized the limitations of conventional regulatory approaches in governing transnational digital platforms operating through algorithmic coordination systems and data-intensive market architectures. Multiple institutional reports highlighted growing concern regarding algorithmic opacity, platform concentration, creator dependency, data asymmetry, and unequal bargaining structures across platform-centered markets. Under these conditions, platform governance increasingly requires adaptive and multi-level coordination mechanisms capable of responding to rapidly evolving technological and market environments.

Recent regulatory developments in Korea illustrate how this triadic interaction operates in practice. The Basic Act on Artificial Intelligence

and Creation of a Trust Base, signed into law in January 2025, introduced obligations covering user notification of AI and AI-generated content, impact assessments for high-impact AI, and transparency about training data and system operations, thereby reshaping the conditions under which AI-augmented content production proceeds. In parallel, the Korea Copyright Commission issued multiple guidelines between 2025 and early 2026 on generative AI outputs, copyright registration, and fair-use considerations for AI training (Korea Copyright Commission, 2025), while MCST signaled forthcoming legislation on text-and-data-mining exceptions and publicity rights. The Korea Communications Commission's Guidelines on the Protection of Users of Generative AI Services (effective March 2025) added a further layer governing platform-user interactions. Together, these instruments demonstrate that policy actors are increasingly intervening at the intersection of IP and platform governance in an attempt to stabilize ecosystem interactions under conditions of compounded technological and institutional uncertainty.

7. Discussion

7.1. Theoretical implications

The study makes four interrelated theoretical contributions, corresponding to the four elements of the framework. First, it extends innovation ecosystem theory beyond its technological and manufacturing foundations by incorporating the distinctive characteristics of content industries, particularly the centrality of intangible assets, symbolic value, and recursive intellectual property circulation. Existing ecosystem frameworks have predominantly conceptualized innovation in terms of technological complementarities and modular product architectures (Adner, 2017; Jacobides et al., 2018). In contrast, the Korean illustration suggests that value creation in digital cultural industries is recursively stabilized through circulating IP under conditions of volatile attention economies, platform-mediated competition, and rapidly shifting audience demand. By positioning IP as the core structural element of the ecosystem, the study shifts the analytical focus from product-based complementarities toward dynamic and interpretive value creation processes organized through intangible and continuously recombinable assets (Caves, 2000; Towse, 2010).

Second, the study advances a process-oriented and mechanism-based understanding of innovation by introducing IP circulation as a central generative mechanism of ecosystem evolution, together with the contingency that governs it. While prior research has emphasized coordination among actors and complementarities across components, it has paid limited attention to how value evolves recursively over time through ongoing recombination, circulation, and adaptive redeployment of intellectual assets. The concept of IP circulation captures how intellectual assets are continuously extended, transformed, and redistributed across platforms, formats, and transnational markets, thereby generating cumulative and path-dependent ecosystem value beyond initial production (Jenkins, 2006; Teece, 2007). The key theoretical move, however, is to specify that this mechanism is not intrinsically value-generating: whether IP circulation produces cumulative value for the originating ecosystem or reproduces dependency depends on the allocation of IP ownership. This contingency distinguishes IP circulation from transmedia storytelling and OSMU, which describe the fact of cross-format movement but do not theorize the conditions under which such movement accumulates or leaks value.

Third, the study contributes to platform ecosystem research by reconceptualizing platforms not merely as intermediaries enabling multi-sided interactions, but as infrastructural governors that organize visibility, monetization, audience access, and ecosystem coordination through algorithmically mediated systems (Srnicek, 2017; Van Dijck et al., 2018). While existing research has emphasized the coordinating role of platforms (Gawer & Cusumano, 2014; Parker et al., 2016), the illustration indicates that platforms simultaneously function as sources

of infrastructural dependency, governance asymmetry, and recursive instability within digitally transforming ecosystems. By distinguishing platform coordination (vertical, infrastructural, gatekept) from network integration (horizontal, relational, peer-based), and by embedding both within a broader framework linking IP circulation and policy orchestration, the study clarifies a distinction that prior accounts have often left implicit and deepens theoretical understanding of how platform-centered ecosystems evolve through contested interactions among infrastructural power, data-driven coordination, and uneven value appropriation.

Fourth, the study advances ecosystem governance theory by introducing policy orchestration as a higher-order mechanism of adaptive stabilization, thereby extending the orchestration construct to a new domain. Although prior ecosystem research has acknowledged the role of institutions and public policy in shaping innovation environments (Autio & Thomas, 2014; Mazzucato, 2018), and although developmental-state and Korean Wave scholarship documents active state coordination of cultural industries (Jin, 2024; Lee & Lee, 2024), a mechanism-level account of how public actors orchestrate platform-mediated content ecosystems has been lacking. Drawing on the orchestration construct in innovation-network research (Dhanaraj & Parkhe, 2006) and governance research (Abbott et al., 2015; Abbott & Snidal, 2010), this study conceptualizes the state as an ecosystem orchestrator that seeks to align actors, resources, infrastructures, and incentives through indirect, distributed, and adaptive mechanisms. Consistent with that construct, orchestration is theorized not as the achievement of systemic equilibrium but as a set of attempts, working through intermediaries the orchestrator does not fully control, to manage fragmentation, asymmetry, infrastructural dependency, and governance tensions. Crucially, this orchestration is analytically divided between two functions: pooling calculable risk through policy finance and absorbing irreducible uncertainty through adaptive, iterative coordination. The Korean case suggests that such orchestration becomes increasingly necessary, though not necessarily sufficient, as ecosystems grow more platform-dependent, uncertainty-intensive, and structurally fragmented.

7.2. Implications for industry dynamics

The findings carry three implications for industry dynamics. First, they highlight the growing importance of platform-oriented strategies as a central determinant of firm competitiveness. As platforms increasingly function as infrastructural governors coordinating production, distribution, visibility, monetization, and audience access, firms must align their strategic behavior with platform logics, including algorithmic visibility systems, engagement-based metrics, data-driven recommendation structures, and platform-specific audience analytics (Nieborg & Poell, 2018; Srnicek, 2017). The Korean illustration repeatedly revealed that firms increasingly compete not only through the intrinsic quality of content, but through their ability to secure advantageous positions within platform-mediated environments. This reflects a broader transformation from product-based competition toward positional and infrastructural competition within ecosystem architectures (Kenney & Zysman, 2016). Strategic partnerships with OTT platforms, co-production arrangements, platform-exclusive distribution contracts, and integrated IP expansion strategies therefore become critical mechanisms for securing scalability and global market access. The ownership contingency, however, implies that such partnerships carry a structural risk: where platform access is obtained by ceding IP ownership, short-term scalability can come at the cost of long-term value retention.

Second, the analysis reveals a broader restructuring of industrial organization characterized by the transition from linear value chains toward networked, platform-centric, and recursively interconnected ecosystem configurations. Traditional distinctions between production, distribution, marketing, and consumption become increasingly blurred as platforms integrate multiple stages of the value creation process

within unified digital infrastructures (Gawer & Cusumano, 2014; Parker et al., 2016). The illustration repeatedly demonstrated that the Korean content ecosystem increasingly operates through interconnected networks linking streaming platforms, webtoons, gaming, music, merchandising, immersive media, talent agencies, production studios, and digital distribution systems. Under these conditions, firms increasingly adopt portfolio-oriented and ecosystem-oriented strategies focused on developing scalable IP assets capable of recursive circulation across multiple formats and transnational markets (Teece, 2007).

Finally, the model underscores the growing importance of ecosystem-level coordination and adaptive governance in shaping industry outcomes. As value creation becomes increasingly interdependent and recursively interconnected, firms can no longer rely solely on internal capabilities or isolated market strategies, but must actively engage in collaborative coordination across platforms, networks, policy systems, and institutional environments (Adner, 2017). This includes adapting to platform governance structures, participating in networked production systems, managing cross-sector partnerships, and responding to evolving regulatory frameworks concerning data governance, AI-driven content production, platform transparency, and intellectual property circulation. Importantly, the illustration suggests that industry dynamics are increasingly shaped not only by market competition, but also by the degree of adaptive alignment achieved across ecosystem components under conditions of uncertainty-intensive digital transformation.

7.3. Policy implications

Three policy implications follow from the analysis. First, the findings highlight the need to move from support-based policy toward orchestration-based governance. Traditional policy approaches in cultural and creative industries have focused primarily on alleviating market failures through subsidies, tax incentives, infrastructure provision, and financial assistance programs (Rodrik, 2004). While such instruments remain important, the illustration suggests that they are insufficient on their own to address the systemic coordination challenges emerging within platform-mediated innovation ecosystems. As the Korean case indicates, value creation increasingly depends on recursive interactions among IP systems, digital platforms, networks, and institutional actors, all of which require alignment across multiple layers of the ecosystem. Consequently, policy would need to move beyond isolated interventions toward governance mechanisms capable of facilitating coordination, reducing fragmentation, strengthening complementarities, and mitigating asymmetrical dependencies among heterogeneous actors (Adner, 2017). This includes supporting intermediary organizations, fostering cross-sector collaboration, facilitating knowledge exchange, and developing institutional infrastructures capable of sustaining ecosystem-level adaptation under volatile digital conditions.

Second, the findings underscore the importance of developing IP-centered policy frameworks that recognize intellectual property not merely as a legal asset, but as a recursively circulating and strategically adaptive resource underpinning ecosystem evolution. Existing policy frameworks have frequently treated IP primarily in terms of ownership protection and enforcement mechanisms. However, the illustration suggests that IP increasingly functions as a dynamic ecosystem resource continuously recombined, extended, and redistributed across platforms, formats, and transnational markets (Jenkins, 2006; Towse, 2010). Given the ownership contingency identified in this study, an especially consequential policy lever concerns where IP ownership is allocated in platform-financed productions. Recent Korean initiatives that condition public finance on the preservation of domestic IP ownership, alongside ongoing legislative discussion of an additional compensation right for audiovisual content, illustrate precisely this lever, and suggest that policy can shape not only the volume of circulation but also whether circulation accumulates or leaks value.

Finally, the study suggests that policy effectiveness increasingly depends on ecosystem-level alignment rather than isolated intervention targeting individual firms or sectors. Digital content ecosystems are structurally interconnected systems in which innovation emerges through recursive interactions among creators, platforms, audiences, institutions, technologies, and financial structures (Autio & Thomas, 2014). As the Korean case indicates, ecosystem sustainability depends not only on firm competitiveness, but also on the degree of adaptive coordination achieved across these interconnected components under conditions of uncertainty-intensive transformation. Consequently, policy interventions would benefit from pursuing system-wide resilience, inclusiveness, and adaptive capacity rather than focusing solely on short-term efficiency or isolated market correction.

7.4. Generalizability and limitations

While this study draws on the Korean content industry, its primary objective is not to explain the Korean case alone, but to develop a theoretically grounded framework for understanding innovation ecosystems operating under conditions of uncertainty-intensive digital transformation. Accordingly, the generalizability of the framework should be understood in terms of analytical generalization rather than statistical representativeness (Eisenhardt & Graebner, 2007; Yin, 2018). The value of the Korean case lies not in its typicality or universal representativeness, but in its analytical capacity to reveal underlying mechanisms, governance tensions, and structural dynamics that become particularly visible within rapidly platformizing and digitally transforming cultural industries. Because the framework is illustrated rather than tested, its central propositions, in particular the ownership contingency governing IP circulation and the division of orchestration labor between calculable risk and irreducible uncertainty, are advanced as theoretically grounded conjectures that invite comparative and longitudinal examination rather than as empirically confirmed regularities.

8. Conclusion

This study has developed a conceptual framework for understanding innovation ecosystems in digital content industries under conditions of uncertainty-intensive digital transformation. Addressing the limitations of existing approaches predominantly grounded in manufacturing and technology-driven contexts, the study reconceptualizes content innovation ecosystems as recursively evolving systems shaped through contested interactions among intellectual property (IP), digital platforms, networks, and adaptive governance mechanisms (Adner, 2017; Jacobides et al., 2018; Nieborg & Poell, 2018). By integrating insights from innovation ecosystem theory, platform economy research, and cultural and creative industries (CCIs) literature, the analysis shifts the focus of ecosystem research away from firm-centric and equilibrium-oriented assumptions toward a process-oriented perspective emphasizing recursive instability, infrastructural dependency, adaptive coordination, and governance complexity under platform-mediated digital conditions. The framework specifies three generative mechanisms, IP circulation, platform coordination, and network integration, coordinated by a higher-order layer of policy orchestration whose two functions correspond to Knight's calculable risk and irreducible uncertainty, and it identifies IP ownership allocation as the contingency that determines whether the ecosystem accumulates value or reproduces dependency.

Several limitations and avenues for future research follow from this approach. Because the framework was developed through conceptual synthesis of prior literature and illustrated through a single revelatory case rather than tested, its propositions remain to be examined comparatively and longitudinally. Its scope is most directly applicable to digital cultural sectors operating under high platform dependency and policy-mediated coordination. Comparative cross-national analysis, longitudinal tracing of ecosystem trajectories, and quantitative testing

of the proposed mechanisms and of the ownership contingency would further extend the framework's empirical and theoretical reach. Despite these limitations, the study contributes to a growing body of scholarship seeking to reconceptualize innovation under conditions of digital and platform-based transformation, offering a theoretically grounded and empirically informed account of how innovation ecosystems attempt to manage unknowns when neither risk nor uncertainty can be fully eliminated.

Ethics statement

This study did not involve human participants, human data, or animal subjects. Ethics committee approval was therefore not required.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The author thanks the editors and anonymous reviewers for their constructive feedback, and would like to express his deepest gratitude to Researcher Dongwoo Kang of the Open Government & Innovation Lab at the KDI School of Public Policy and Management for his invaluable administrative support and assistance in carrying out this research.

Data availability

The authors do not have permission to share data.

References

- Abbott, K. W., Genschel, P., Snidal, D., & Zangl, B. (Eds.). (2015). *International organizations as orchestrators*. Cambridge University Press.
- Abbott, K. W., & Snidal, D. (2010). International regulation without international government: Improving IO performance through orchestration. *The Review of International Organizations*, 5(3), 315–344. <https://doi.org/10.1007/s11558-010-9092-3>
- Adner, R. (2017). Ecosystem as structure: An actionable construct for strategy. *Journal of Management*, 43(1), 39–58. <https://doi.org/10.1177/0149206316678451>
- Adner, R., & Kapoor, R. (2010). Value creation in innovation ecosystems: How the structure of technological interdependence affects firm performance in new technology generations. *Strategic Management Journal*, 31(3), 306–333. <https://doi.org/10.1002/smj.821>
- Ansell, C., & Boin, A. (2019). Taming deep uncertainty: The potential of pragmatist principles for understanding and improving strategic crisis management. *Administration & Society*, 51(7), 1079–1112. <https://doi.org/10.1177/0095399717747655>
- Autio, E., Nambisan, S., Thomas, L. D. W., & Wright, M. (2018). Digital affordances, spatial affordances, and the genesis of entrepreneurial ecosystems. *Strategic Entrepreneurship Journal*, 12(1), 72–95. <https://doi.org/10.1002/sej.1266>
- Autio, E., & Thomas, L. D. W. (2014). Innovation ecosystems: Implications for innovation management. In M. Dodgson, D. M. Gann, & N. Phillips (Eds.), *The Oxford handbook of innovation management* (pp. 204–228). Oxford University Press.
- Brem, A., Nylund, P. A., & Roshani, S. (2024). Unpacking the complexities of crisis innovation: A comprehensive review of ecosystem-level responses to exogenous shocks. *Review of Managerial Science*, 18(8), 2441–2464. <https://doi.org/10.1007/s11846-023-00682-5>
- Caves, R. E. (2000). *Creative industries: Contracts between art and commerce*. Harvard University Press.
- Cornelissen, J. P. (2017). Editor's comments: Developing propositions, a process model, or a typology? Addressing the challenges of writing theory without a boilerplate. *Academy of Management Review*, 42(1), 1–9. <https://doi.org/10.5465/amr.2016.0196>
- Cunningham, S., & Craig, D. (2019). *Social media entertainment: The new intersection of Hollywood and Silicon Valley*. NYU Press.
- Cunningham, S., & Potts, J. (2015). Creative industries and the wider economy. In C. Jones, M. Lorenzen, & J. Sapsed (Eds.), *The Oxford handbook of creative industries* (pp. 387–404). Oxford University Press.
- DeFillippi, R., Grabher, G., & Jones, C. (2007). Introduction to paradoxes of creativity: Managerial and organizational challenges in the cultural economy. *Journal of Organizational Behavior*, 28(5), 511–521. <https://doi.org/10.1002/job.466>
- DeFillippi, R. J., & Arthur, M. B. (1998). Paradox in project-based enterprise: The case of film making. *California Management Review*, 40(2), 125–139. <https://doi.org/10.2307/41165936>
- Dhanaraj, C., & Parkhe, A. (2006). Orchestrating innovation networks. *Academy of Management Review*, 31(3), 659–669. <https://doi.org/10.5465/amr.2006.21318923>
- Dosi, G. (1982). Technological paradigms and technological trajectories: A suggested interpretation of the determinants and directions of technical change. *Research Policy*, 11(3), 147–162. [https://doi.org/10.1016/0048-7333\(82\)90016-6](https://doi.org/10.1016/0048-7333(82)90016-6)
- Eisenhardt, K. M., & Graebner, M. E. (2007). Theory building from cases: Opportunities and challenges. *Academy of Management Journal*, 50(1), 25–32. <https://doi.org/10.5465/amj.2007.24160888>
- Evans, D. S., & Schmalensee, R. (2016). *Matchmakers: The new economics of multisided platforms*. Harvard Business Review Press.
- Gans, J. S., Hsu, D. H., & Stern, S. (2002). When does start-up innovation spur the gale of creative destruction? *The RAND Journal of Economics*, 33(4), 571–586. <https://doi.org/10.2307/3087475>
- Gawer, A. (2014). Bridging differing perspectives on technological platforms: Toward an integrative framework. *Research Policy*, 43(7), 1239–1249. <https://doi.org/10.1016/j.respol.2014.03.006>
- Gawer, A. (2021). Digital platforms' boundaries: The interplay of firm scope, platform sides, and digital interfaces. *Long Range Planning*, 54(5), Article 102045. <https://doi.org/10.1016/j.lrp.2020.102045>
- Gawer, A., & Cusumano, M. A. (2014). Industry platforms and ecosystem innovation. *Journal of Product Innovation Management*, 31(3), 417–433. <https://doi.org/10.1111/jipm.12105>
- Gillespie, T. (2018). *Custodians of the internet: Platforms, content moderation, and the hidden decisions that shape social media*. Yale University Press.
- Grabher, G. (2002). Cool projects, boring institutions: Temporary collaboration in social context. *Regional Studies*, 36(3), 205–214. <https://doi.org/10.1080/00343400220122025>
- Granovetter, M. (1985). Economic action and social structure: The problem of embeddedness. *American Journal of Sociology*, 91(3), 481–510. <https://doi.org/10.1086/228311>
- Hesmondhalgh, D. (2019). *The cultural industries* (4th ed.). Sage.
- Hirsch, P. M. (1972). Processing fads and fashions: An organization-set analysis of cultural industry systems. *American Journal of Sociology*, 77(4), 639–659. <https://doi.org/10.1086/225192>
- Jaakkola, E. (2020). Designing conceptual articles: Four approaches. *AMS Review*, 10(1–2), 18–26. <https://doi.org/10.1007/s13162-020-00161-0>
- Jacobides, M. G., Cennamo, C., & Gawer, A. (2018). Towards a theory of ecosystems. *Strategic Management Journal*, 39(8), 2255–2276. <https://doi.org/10.1002/smj.2904>
- Jenkins, H. (2006). *Convergence culture: Where old and new media collide*. NYU Press.
- Jin, D. Y. (2023). *Understanding Korean webtoon culture: Transmedia storytelling, digital platforms, and genres*. Harvard University Asia Center.
- Jin, D. Y. (2024). The rise of digital platforms as a soft power apparatus in the New Korean Wave era. *Communication and the Public*, 9(2), 161–177. <https://doi.org/10.1177/20570473241234204>
- Kattel, R., & Mazzucato, M. (2018). Mission-oriented innovation policy and dynamic capabilities in the public sector. *Industrial and Corporate Change*, 27(5), 787–801. <https://doi.org/10.1093/icc/dty032>
- Kenney, M., & Zysman, J. (2016). The rise of the platform economy. *Issues in Science & Technology*, 32(3), 61–69.
- Khizar, H. M. U., Iqbal, M. J., Khalid, J., & Adomako, S. (2025). Digital technologies in innovation ecosystem: A systematic review of current trends and future perspective. *R&D Management*. <https://doi.org/10.1111/radm.12758>. Advance online publication.
- Kim, T. (2022). Cultural politics of Netflix in local contexts: A case of the Korean media industries. *Media, Culture & Society*, 44(7), 1391–1406. <https://doi.org/10.1177/01634437221111917>
- Kim, W. (2025). Reconfiguring the global-local nexus: A comparative analysis of Netflix and theatrical film diversity in South Korea. *Convergence: The International Journal of Research Into New Media Technologies*. <https://doi.org/10.1177/13548565251393826>. Advance online publication.
- Knight, F. H. (1921). *Risk, uncertainty and profit*. Houghton Mifflin.
- Kogut, B. (2000). The network as knowledge: Generative rules and the emergence of structure. *Strategic Management Journal*, 21(3), 405–425. [https://doi.org/10.1002/\(SICI\)1097-0266\(200003\)21:3<3C405::AID-SMJ103>3E3.0.CO;2-5](https://doi.org/10.1002/(SICI)1097-0266(200003)21:3<3C405::AID-SMJ103>3E3.0.CO;2-5)
- Korea Copyright Commission. (2025). *Policy guidelines on generative AI and copyright issues*.
- Korea Creative Content Agency [KOCCA]. (2024). *2023 content industry white paper [2023 콘텐츠산업 백서]*. Korea Creative Content Agency.
- Korea Creative Content Agency [KOCCA]. (2025). *2024 content IP transaction status survey [2024 콘텐츠 IP 거래 현황 조사]*. Korea Creative Content Agency.
- Korea Development Institute [KDI]. (2024). *Empirical study on the growth of Korea's cultural content industry and policy implications [문화콘텐츠산업의 성장에 관한 실증적 고찰과 정책 방향]* (Research Monograph 2024-02; J. Lee). Korea Development Institute. <https://doi.org/10.22740/kdi.rm.2024.02>
- Korea Information Society Development Institute [KISDI]. (2025). *KISDI media statistics report on broadcasting and OTT markets*. Korea Information Society Development Institute.

- Korean Film Council [KOFIC]. (2025). *2024 Korean film industry annual report [2024년 한국 영화산업 결산] (KOFIC Research 2025-02)*. Korean Film Council.
- Lampel, J., Lant, T., & Shamsie, J. (2000). Balancing act: Learning from organizing practices in cultural industries. *Organization Science*, 11(3), 263–269. <https://doi.org/10.1287/orsc.11.3.263.12503>
- Lee, S.-K., & Lee, S.-M. (2024). A study on the policy background of the formation of Korean-style content intellectual property (IP) industry: Focusing on the animation-character industry policy [한국형 콘텐츠 지식재산(IP) 산업 형성의 정책적 배경에 대한 연구]. *Journal of Social Science [사회과학연구]*, 63(2), 205–249. <https://doi.org/10.22418/JSS.2024.8.63.2.205>
- Lee, Y. (2024). A study on OTT use of domestic webtoons and worldview expansion of webtoon intellectual property (IP) [국내 웹툰 원작의 OTT 활용 사례와 웹툰 지식재산(IP)의 세계관 확장에 대한 제언]. *Korea and Global Affairs [한국과 세계]*, 6(2), 187–216. <https://doi.org/10.22743/kwr.2024.6.2.187>
- Lessig, L. (2008). *Remix: Making art and commerce thrive in the hybrid economy*. Penguin Press.
- Mazzucato, M. (2018). Mission-oriented innovation policies: Challenges and opportunities. *Industrial and Corporate Change*, 27(5), 803–815. <https://doi.org/10.1093/icc/dty034>
- Ministry of Culture, Sports and Tourism [MCST]. (2025a). *2025 major business plan [2025년 주요 업무 추진계획]*. Ministry of Culture, Sports and Tourism.
- Ministry of Culture, Sports and Tourism [MCST]. (2025b). *2026 major business plan [2026년 문화체육관광부 주요 업무계획]*. Ministry of Culture, Sports and Tourism.
- Ministry of Culture, Sports and Tourism [MCST]. (2026). *2025 content industry survey results (2024 reference year) [2025년 콘텐츠산업조사 결과]*. Ministry of Culture, Sports and Tourism.
- Nambisan, S., Lyytinen, K., Majchrzak, A., & Song, M. (2017). Digital innovation management: Reinventing innovation management research in a digital world. *MIS Quarterly*, 41(1), 223–238. <https://doi.org/10.25300/MISQ/2017/41.1.03>
- Nambisan, S., Wright, M., & Feldman, M. (2019). The digital transformation of innovation and entrepreneurship: Progress, challenges and key themes. *Research Policy*, 48(8), Article 103773. <https://doi.org/10.1016/j.respol.2019.03.018>
- Nelson, R. R., & Winter, S. G. (1982). *An evolutionary theory of economic change*. Harvard University Press.
- Nieborg, D. B., & Poell, T. (2018). The platformization of cultural production: Theorizing the contingent cultural commodity. *New Media & Society*, 20(11), 4275–4292. <https://doi.org/10.1177/1461444818769694>
- Parker, G. G., Van Alstyne, M. W., & Choudary, S. P. (2016). *Platform revolution: How networked markets are transforming the economy—and how to make them work for you*. W. W. Norton.
- Patton, M. Q. (2014). *Qualitative research and evaluation methods: Integrating theory and practice* (4th ed.). Sage.
- Pisano, G. P. (2006). Profiting from innovation and the intellectual property revolution. *Research Policy*, 35(8), 1122–1130. <https://doi.org/10.1016/j.respol.2006.09.008>
- Plantin, J.-C., Lagoze, C., Edwards, P. N., & Sandvig, C. (2018). Infrastructure studies meet platform studies in the age of Google and Facebook. *New Media & Society*, 20(1), 293–310. <https://doi.org/10.1177/1461444816661553>
- Poell, T., Nieborg, D. B., & Duffy, B. E. (2022). *Platforms and cultural production*. Polity.
- Potts, J., Cunningham, S., Hartley, J., & Ormerod, P. (2008). Social network markets: A new definition of the creative industries. *Journal of Cultural Economics*, 32(3), 167–185. <https://doi.org/10.1007/s10824-008-9066-y>
- Powell, W. W., Koput, K. W., & Smith-Doerr, L. (1996). Interorganizational collaboration and the locus of innovation: Networks of learning in biotechnology. *Administrative Science Quarterly*, 41(1), 116–145. <https://doi.org/10.2307/2393988>
- Pyo, J. Y., Jang, M., & Yoon, T.-J. (2019). Dynamics between agents in the new webtoon ecosystem in Korea: Responses to waves of transmedia and transnationalism. *International Journal of Communication*, 13, 2161–2178.
- Rochet, J.-C., & Tirole, J. (2003). Platform competition in two-sided markets. *Journal of the European Economic Association*, 1(4), 990–1029. <https://doi.org/10.1162/154247603322493212>
- Rodrik, D. (2004). *Industrial policy for the twenty-first century (CEPR Discussion Paper No. 4767)*. Centre for Economic Policy Research.
- Shapiro, C., & Varian, H. R. (1999). *Information rules: A strategic guide to the network economy*. Harvard Business School Press.
- Siggelkow, N. (2007). Persuasion with case studies. *Academy of Management Journal*, 50(1), 20–24. <https://doi.org/10.5465/amj.2007.24160882>
- Srnicek, N. (2017). *Platform capitalism*. Polity.
- Teece, D. J. (1986). Profiting from technological innovation: Implications for integration, collaboration, licensing and public policy. *Research Policy*, 15(6), 285–305. [https://doi.org/10.1016/0048-7333\(86\)90027-2](https://doi.org/10.1016/0048-7333(86)90027-2)
- Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal*, 28(13), 1319–1350. <https://doi.org/10.1002/smj.640>
- Teece, D. J., Peteraf, M., & Leih, S. (2016). Dynamic capabilities and organizational agility: Risk, uncertainty, and strategy in the innovation economy. *California Management Review*, 58(4), 13–35. <https://doi.org/10.1525/cmr.2016.58.4.13>
- Throsby, D. (2001). *Economics and culture*. Cambridge University Press.
- Tiwana, A. (2014). *Platform ecosystems: Aligning architecture, governance, and strategy*. Morgan Kaufmann.
- Towse, R. (2010). *A textbook of cultural economics*. Cambridge University Press.
- Van Dijk, J., Poell, T., & De Waal, M. (2018). *The platform society: Public values in a connective world*. Oxford University Press.
- Yin, R. K. (2018). *Case study research and applications: Design and methods* (6th ed.). Sage.
- Zuboff, S. (2019). *The age of surveillance capitalism: The fight for a human future at the new frontier of power*. PublicAffairs.